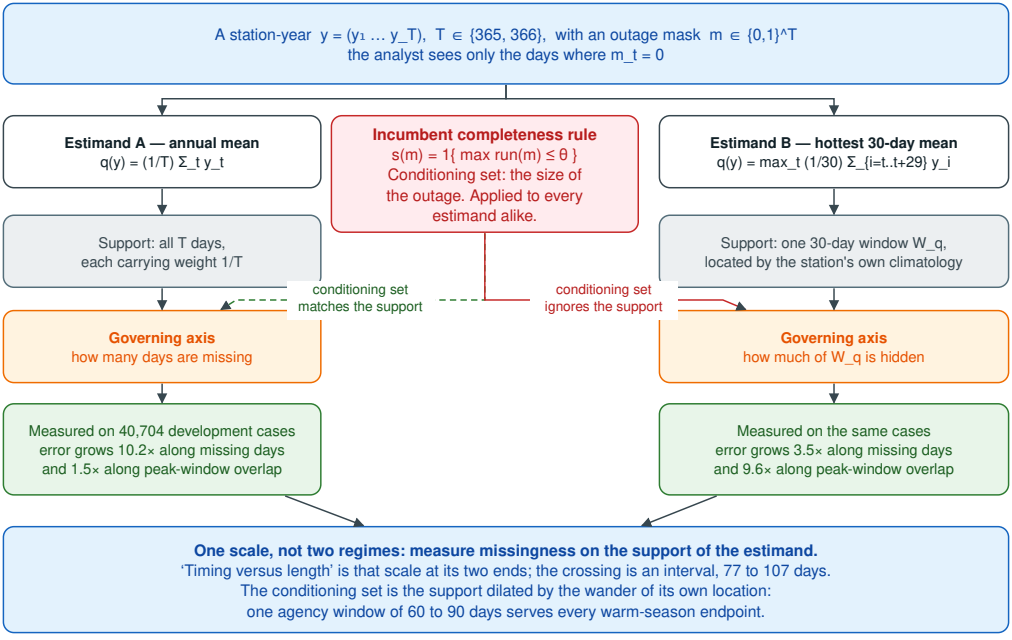
A station-year y = (y₁ … y_T), T ∈ {365, 366}, with an outage mask m ∈ {0,1}^T
the analyst sees only the days where m_t = 0
Estimand A — annual mean
q(y) = (1/T) Σ_t y_t
Incumbent completeness rule
s(m) = 1{ max run(m) ≤ θ }
Conditioning set: the size of the outage. Applied to every estimand alike.
Estimand B — hottest 30-day mean
q(y) = max_t (1/30) Σ_{i=t..t+29} y_i
Support: all T days, each carrying weight 1/T
Support: one 30-day window W_q, located by the station's own climatology
conditioning set matches the support
conditioning set ignores the support
Governing axis
how many days are missing
Governing axis
how much of W_q is hidden
Measured on 40,704 development cases
error grows 10.2× along missing days
and 1.5× along peak-window overlap
Measured on the same cases
error grows 3.5× along missing days
and 9.6× along peak-window overlap
One scale, not two regimes: measure missingness on the support of the estimand.
'Timing versus length' is that scale at its two ends; the crossing is an interval, 77 to 107 days.
The conditioning set is the support dilated by the wander of its own location:
one agency window of 60 to 90 days serves every warm-season endpoint.